2-to-1 Binomials on Finite Fields of Odd Characteristic

Lucas Krompholz, Gohar Kyureghyan, Alexander Oertel

University of Rostock

WCC Paris, June 2026

The Beginning

Binomials in *EVEN* characteristic:

Let $b(x) = x^s + ux^t$ with $u \in \mathbb{F}_{2^n}^*$. The binomial is called 2-to-1 if

$$\left| b^{-1}(\{z\}) \right| \in \{0, 2\}$$

for all $z \in \mathbb{F}_{2^n}$.

The Beginning

Theorem (Kölsch and Kyureghyan, 2025)

Let $b(x) = x^s + ux^t$ be a 2-to-1 binomial on $\mathbb{F}_{2^n}$ for some $u \in \mathbb{F}_{2^n}^*$, then $gcd(s-t, 2^n-1) = gcd(s, 2^n-1) = gcd(t, 2^n-1) = 1$.

The following statements are equivalent:

- There is one $u \in \mathbb{F}_{2^n}^*$ such that the binomial $b(x) = x^s + ux^t$ is 2-to-1 on $\mathbb{F}_{2^n}$
- For every $u \in \mathbb{F}_{2^n}^*$ the binomial $b(x) = x^s + ux^t$ is 2-to-1 on $\mathbb{F}_{2^n}$
- $x^{s/t}$ is an o-monomial on $\mathbb{F}_{2^n}$

What can be said about binomials in odd characteristic?

What can be said about binomials in odd characteristic?

Challenges:

- Can we control the preimages of arbitrary binomials?
- How to define 2-to-1?

Some notation

We fix the following notation

$$b(x) = x^s - ux^t,$$

where $1 \leq t < s \leq q-2$.

Additionally, we use

- $d = gcd(s-t, q-1)$
- $d' = gcd(s, t, q-1)$ (so $d'|d$)
- $U_d = \{x \in \mathbb{F}_q^* : x^d = 1\}$, the subgroup of size d
- $\mu_d = \{x^d : x \in \mathbb{F}_q^*\}$, the subgroup of size $(q-1)/d$

The main idea to control preimages

We want to find solutions $x, y \in \mathbb{F}_q^*$ such that $x \neq y$, and

$$b(x) = b(y).$$

The main idea to control preimages

We want to find solutions $x, y \in \mathbb{F}_q^*$ such that $x \neq y$, and

$$b(x) = b(y).$$

We use the following ansatz: Find all $\alpha \in \mathbb{F}_q^*$ such that

$$b(y) = b(\alpha y).$$

For which α do we have a solution?

Let $y \in \mathbb{F}_q^*$.

$$\begin{aligned}
\iff \quad & b(y) = b(\alpha y) \\
\iff \quad & y^s - uy^t = \alpha^s y^s - u\alpha^t y^t \\
\iff \quad & y^s(\alpha^s - 1) = uy^t(\alpha^t - 1) \\
\iff \quad & u^{-1} \cdot y^{s-t} = \frac{\alpha^t - 1}{\alpha^s - 1}
\end{aligned}$$

If $\alpha^s - 1 = 0$, then also $\alpha^t - 1 = 0$.
$\implies \alpha \in U_{d'}$. This is a trivial solution.

The set of non-trivial α

For fixed $y \in \mathbb{F}_q^*$ define

$$\begin{aligned}\mathcal{A}(y) &:= \left\{\alpha \in \mathbb{F}_q^* \setminus (U_s \cup U_t) : b(y) = b(\alpha y)\right\} \\ &= \left\{\alpha \in \mathbb{F}_q^* \setminus (U_s \cup U_t) : u^{-1} y^{s-t} = \frac{\alpha^t - 1}{\alpha^s - 1}\right\}.\end{aligned}$$

Combining the solutions

Theorem 1

Let $y \in \mathbb{F}_q^*$ be such that $z := b(y) \neq 0$. Then $b(\alpha y) = b(y)$ if and only if $\alpha \in \mathcal{A}(y) \cup U_{d'}$, where $d' = \gcd(s, t, q-1)$. In particular,

$$\left| b^{-1}(\{z\}) \right| = |\mathcal{A}(y)| + d'.$$

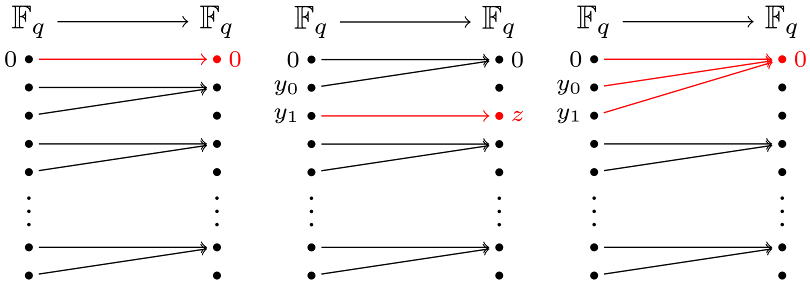
Definition of 2-to-1

Let q be odd and $f : \mathbb{F}_q \to \mathbb{F}_q$ with $f(0) = 0$.

Define the following types of 2-to-1:

- **type 1:** each non-zero element in its image set has exactly two non-zero preimages and $f(x) = 0$ only for $x = 0$.
- **type 2:** there is a unique $z \neq 0$ in its image set with exactly one preimage and the remaining elements from the image set of f have exactly two preimages.
- **type 3:** each non-zero element in its image set has exactly two preimages and 0 has three preimages.

Visualization of different types

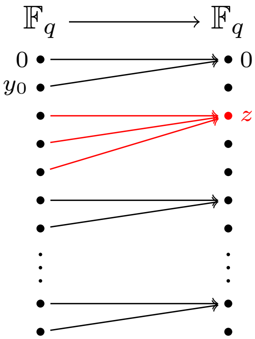


(a) 2-to-1 map of type 1 (b) 2-to-1 map of type 2 (c) 2-to-1 map of type 3

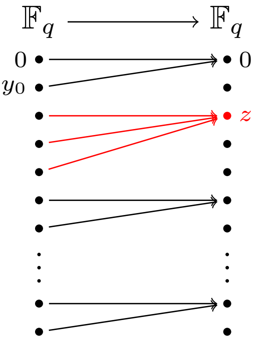
We call the element $z \in \mathbb{F}_q$ in the image with $|b^{-1}(z)| \neq 0, 2$ the *exceptional* element.

Each 2-to-1 map has exactly one exceptional element.

Why only these types?



Why only these types?



Proposition

Suppose the exceptional element $z \in \text{Image}(b)$ is non-zero such that $z' \in \text{Image}(b) \setminus \{z\}$ has exactly two preimages under b. Then z has exactly one preimage. In particular, $b(x)$ is a 2-to-1 binomial of type 2.

Applying Theorem 1 to 2-to-1 binomials

Recall our theorem:

Theorem 1

Let $y \in \mathbb{F}_q^*$ be such that $z := b(y)$ is non-zero. Then $b(\alpha y) = b(y)$ if and only if $\alpha \in \mathcal{A}(y) \cup U_{d'}$, where $d' = \gcd(s, t, q-1)$. In particular,

$$\left| b^{-1}(\{z\}) \right| = |\mathcal{A}(y)| + d'.$$

Applying this to the different types yields:

Type 1

$b(x)$ is type 1 if and only if $\gcd(s-t, q-1) = d \geq 2$, $u \notin \mu_d$ and $|\mathcal{A}(y)| + \gcd(s, t, q-1) = 2$ for all $y \in \mathbb{F}_q^*$.

Type 2

There are non-zero y_0, y_1 such that $b(y_0) = b(0) = 0$ and $b(y_1)$ exceptional if and only if
$\gcd(s-t, q-1) = 1$ and $|\mathcal{A}(y)| = 1$ for all $y \in \mathbb{F}_q \setminus \{y_0, y_1\}$.

Note that in this case any parameter $u \neq 0$ defines a 2-to-1 binomial of type 2.

Applying Theorem 1 to 2-to-1 binomials

Type 3

The binomial $b(x)$ is 2-to-1 of type 3 if and only if $\gcd(s-t, q-1) = 2$, $u \in \mu_2$, and for every $y \in \mathbb{F}_q^*$

$$|\mathcal{A}(y)| = \begin{cases} 0 & \text{if } \gcd(s,t,q-1) = 2 \\ 1 & \text{if } \gcd(s,t,q-1) = 1. \end{cases}$$

Some simple examples

Binomial	q	u	type
$b(x) = perm(x^2)$	any	-	1
$x^{\frac{q+1}{2}} - ux$	any	(u-1)(u+1) non-square	1
$x^2 - ux$	any	$u \neq 0$	2
$x^{q-2} - ux$	$q \equiv 1 \mod 4$	$-u$ non-square	1
$x^{q-2} - ux$	$q \equiv 3 \mod 4$	$-u$ non-square	3
$x^{\frac{q-1}{2}+1} - ux^{\frac{q-1}{2}-1}$	$q \equiv 3 \mod 4$	u square	3

Idea of the proofs

Consider $b(x) = x^{\frac{q-1}{2}+1} - ux^{\frac{q-1}{2}-1}$.

From $q \equiv 3 \mod 4$ we have $d' = \gcd(\frac{q+1}{2}, \frac{q-3}{2}, q-1) = 2$.

For $y \in \mathbb{F}_q^*$ we need to solve

$$u^{-1}y^2 = \frac{\alpha^{\frac{q-1}{2}} \frac{1}{\alpha} - 1}{\alpha^{\frac{q-1}{2}} \alpha - 1}.$$

- If $\alpha^{\frac{q-1}{2}} = 1$, this equation becomes $u^{-1}y^2 = -\alpha^{-1}$.
  No solutions because -1 is no square since $q \equiv 3 \mod 4$.
- If $\alpha^{\frac{q-1}{2}} = -1$ this equation becomes $u^{-1}y^2 = \alpha^{-1}$.
  No solutions either.

Towards a characterization of type 2 binomials

Lemma

A 2-to-1 binomial $b(x) = x^s - ux^t$ of type 2 on $\mathbb{F}_q$ satisfies either

$$\gcd(s, q-1) = 2 \text{ and } \gcd(t, q-1) = 1$$

or

$$\gcd(s, q-1) = 1 \text{ and } \gcd(t, q-1) = 2.$$

Definition of ovals

$\mathcal{O} \subset \mathrm{PG}(2,q)$ is an oval if

- $|\mathcal{O}| = q + 1$
- for each line l of $\mathrm{PG}(2,q)$, $|l \cap \mathcal{O}| \leq 2$.

Complete characterization of ovals in odd characteristic:

Theorem (Segre's Theorem)

If q is odd, then any oval of $\mathrm{PG}(2,q)$ is a conic.

Connection to ovals

Lemma

Let q be odd and $b(x) = x^s + x^t$ be 2-to-1 of type 2 with $\gcd(t, q-1) = 1$. Then the set

$$\mathcal{O}(s,t) := \{(1, y^t, y^s) : y \in \mathbb{F}_q\} \cup \{(0,0,1)\} \subset \mathrm{PG}(2,q)$$

is an oval.

Idea of the proof

First, note that $|\mathcal{O}(s,t)| = q+1$.

Let $l_{a,u} := \left\langle \left(\begin{smallmatrix} a \\ u \\ 1 \end{smallmatrix}\right) \right\rangle^{\perp}$ be a line of $\mathrm{PG}(2,q)$.
$(0,0,1)$ is not on $l_{a,u}$ and $(1, y^t, y^s) \in \mathcal{O}(s,t)$ is on $l_{a,b}$ if and only if

$$0 = (1, y^t, y^s) \cdot \begin{pmatrix} a \\ u \\ 1 \end{pmatrix} = a + uy^t + y^s.$$

There are at most 2 solutions since $uy^t + y^s$ is 2-to-1 of type 2.

The other lines $l_a := \left\langle \left(\begin{smallmatrix} a \\ 1 \\ 0 \end{smallmatrix}\right) \right\rangle^{\perp}$, and $l_\infty := \left\langle \left(\begin{smallmatrix} 1 \\ 0 \\ 0 \end{smallmatrix}\right) \right\rangle^{\perp}$ also contain at most 2 points.

Complete Characterization

Theorem

Let q be odd and let $s, t \in \mathbb{N}$. The binomial

$$b(x) = x^s - ux^t$$

is 2-to-1 of type 2 on $\mathbb{F}_q$ if and only if $u \in \mathbb{F}_q^*$ is arbitrary and the pair of exponents (s, t) satisfies one of the following conditions:

$$\begin{cases} \gcd(t, q-1) = 1 \textit{ and } s \equiv 2t \bmod q-1, \\ \gcd(s, q-1) = 1 \textit{ and } t \equiv 2s \bmod q-1. \end{cases}$$

Any type 2 binomial $b(x)$ can be written as $b(x) = (x^2 - ux) \circ (x^c)$, where $\gcd(c, q-1) = 1$.

Thank you for listening.